Experiment 7:

Code:

```c
#include<stdio.h>

int main(){
long long counter =0;

while(1){
counter++;
if(counter % 100000000 == 0){
printf("CPU task running: %lld\n", counter);
}
}
return 0;

}
```

Output:

```
┌──(kali㉿sourav)-[~]
└─$ nano exp7a.c

┌──(kali㉿sourav)-[~]
└─$ gcc exp7a.c -o exp7a

┌──(kali㉿sourav)-[~]
└─$ ./exp7a
CPU task running: 100000000
CPU task running: 200000000
CPU task running: 300000000
CPU task running: 400000000
CPU task running: 500000000
```

```
┌──(kali㉿sourav)-[~]
└─$ pgrep -a exp7a
17871 ./exp7a

┌──(kali㉿sourav)-[~]
└─$ top -p 17871
```

```
top - 12:15:17 up 32 min,  1 user,  load average: 1.09, 0.64, 0.43
Tasks:   1 total,   1 running,   0 sleeping,   0 stopped,   0 zombie
%Cpu(s): 15.9 us,  0.4 sy,  0.0 ni, 83.6 id,  0.1 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem :   3916.6 total,   2018.9 free,   1204.9 used,    973.1 buff/cache
MiB Swap:    953.7 total,    953.7 free,      0.0 used.   2711.7 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  17871 kali      20   0    2580   1560   1460 R 100.0   0.0   1:32.16 exp7a
```

Experiment 7:

Code:

```c
#include <stdio.h>
#include <unistd.h>
int main(){
FILE *fp;
printf("Starting I/O task ... \n");
while(1){
fp = fopen("dummy.txt", "w");
if(fp != NULL){
fprintf(fp, "Writing disk data\n");
fclose(fp);
printf("File operation done\n");
}
sleep(1);
}
return 0;

}
```

Output:

```
┌──(kali㉿sourav)-[~]
└─$ nano exp7b.c

┌──(kali㉿sourav)-[~]
└─$ gcc exp7b.c -o exp7b

┌──(kali㉿sourav)-[~]
└─$ ./exp7b
Starting I/O task ...
File operation done
File operation done
File operation done
File operation done
File operation done
```

```
┌──(kali㉿sourav)-[~]
└─$ pgrep -a exp7b
21359 ./exp7b

┌──(kali㉿sourav)-[~]
└─$ top -p 21359
```

```
top - 12:23:20 up 40 min,  1 user,  load average: 0.09, 0.24, 0.31
Tasks:   1 total,   0 running,   1 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.1 us,  0.3 sy,  0.0 ni, 99.5 id,  0.0 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem :   3916.6 total,   1996.5 free,   1226.6 used,    972.7 buff/cache
MiB Swap:    953.7 total,    953.7 free,      0.0 used.   2690.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
  21359 kali      20   0    2580   1608   1508 S   0.0   0.0   0:00.14 exp7b
```